

#oms 1.18

##1

$X \sim \Gamma(\lambda, r)$

$f_X(x) = \lambda^r * \Gamma(r)^{-1} * x^{r-1} * \exp(-\lambda * x)$

$\Rightarrow L(\lambda, r) = \prod_{i=1}^n f_X(\lambda, r, x_i) = \lambda^{r*n} * \Gamma(r)^{-n} * \prod_{i=1}^n x_i^{r-1} * \exp(-\lambda \sum_{i=1}^n x_i)$

##2

$l(\lambda, r) = r * n * \log(\lambda) - n * \log(\Gamma(r)) + (r - 1) \sum_{i=1}^n \log(x_i) - \lambda * \sum_{i=1}^n x_i$ $\frac{\partial l}{\partial \lambda} = \frac{r * n}{\lambda} - \sum_{i=1}^n x_i = 0$

$\frac{\partial l}{\partial r} = n * \log(\lambda) - n * \frac{d}{dr} \log(\Gamma(r)) + \sum_{i=1}^n \log(x_i) = 0$

##3

```
set.seed(1)
x <- rgamma(1000, shape = 2, rate = 3)

An<-mean(x)
Bn<-mean(log(x))
f<-function(x){
  return (digamma(x)-log(x)-Bn+log(An))
}

r_hat<-uniroot(f,c(1,100))$root

lambda_hat<-r_hat/An
print("r estimat")
```

```
## [1] "r estimat"
```

```
r_hat
```

```
## [1] 1.875565
```

```
print("lambda estimat")
```

```
## [1] "lambda estimat"
```

```
lambda_hat
```

```
## [1] 2.831249
```